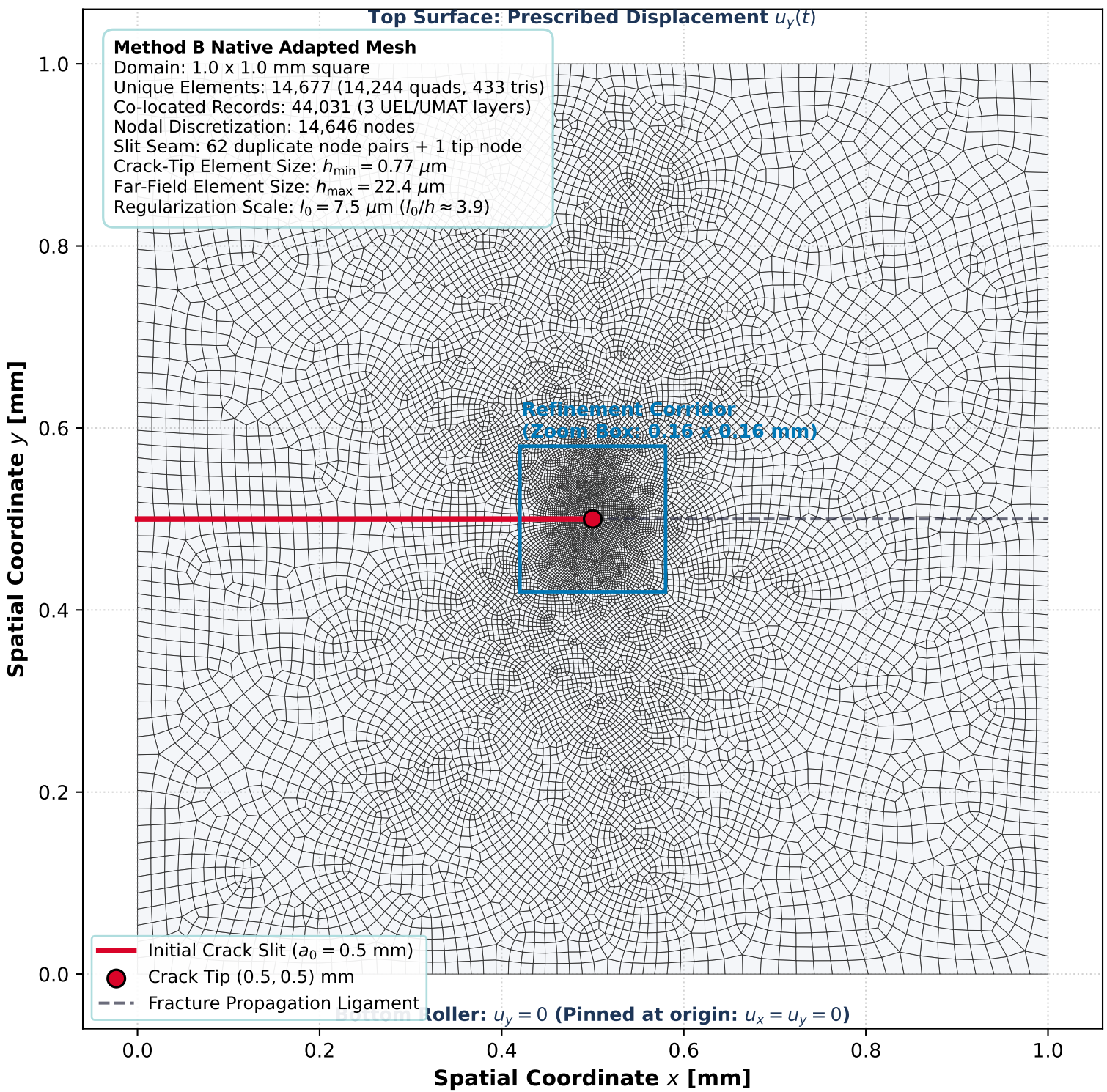

Pandey--Kumar Mode-I Benchmark: Genuine Method B Native Adapted Mesh
(14,677 Elements, Native Abaqus CAE Pre-Peak Elastic Error Refinement)
Top Surface: Prescribed Displacement $u_y(t)$
Method B Native Adapted Mesh
Domain: 1.0 x 1.0 mm square
Unique Elements: 14,677 (14,244 quads, 433 tris)
Co-located Records: 44,031 (3 UEL/UMAT layers)
Nodal Discretization: 14,646 nodes
Slit Seam: 62 duplicate node pairs + 1 tip node
Crack-Tip Element Size: $h_{min} = 0.77$ μm
Far-Field Element Size: $h_{max} = 22.4$ μm
Regularization Scale: $l_0 = 7.5$ μm ($l_0/h \approx 3.9$)
Refinement Corridor
(Zoom Box: 0.16 x 0.16 mm)
Initial Crack Slit ($a_0 = 0.5$ mm)
Crack Tip (0.5, 0.5) mm
Fracture Propagation Ligament
oller: $u_y = 0$ (Pinned at origin: $u_x = u_y = 0$)
Spatial Coordinate y [mm]
Spatial Coordinate x [mm]
0.0
0.2
0.4
0.6
0.8
1.0